

White Paper 10 — Compiling the report and deliverables

Pipeline step: Compile report and deliverables **Inputs:** all upstream artifacts (tables, structures, figures, packages) **Outputs:** `CONSOLIDATED_REPORT.md`, `lead_pipeline_summary.csv`, `fig_pipeline_funnel.png`

1. Purpose

The final step assembles the pipeline's outputs into a coherent, self-documenting deliverable set: a consolidated report, a master summary table, and a pipeline overview figure — each written so that a reader (here, a clinician collaborator) can follow the logic from psoriasis differential expression to cluster-ready simulation packages without reading the code.

2. What the pipeline produced

The pipeline flows through a clear funnel:

- **2,379** significant meta-analysis DEGs (PP vs NN, FDR < 0.05 & $|\log_2FC| > 1$);
- **154** protein-coding candidates annotated for druggability;
- **~28** tractable, on-psoriasis-axis targets;
- **3** structure-ready leads — **STAT3**, **RORC (RORyt)**, **JAK3** — forming a continuous JAK-STAT/Th17 signaling axis;
- **1** target (RORyt) taken through a validated 151-compound docking screen, with all three leads packaged for cluster-scale docking, co-folding, and MD.

3. The deliverable set

Analysis tables: `target_universe.csv`, `druggability_annotation.csv`, `druggable_target_ranking.csv`, `pocket_report.csv`, `docking_ranked.csv`, `lead_pipeline_summary.csv`.

Structures: the three lead structures and the top-hit complex (`RORC_tophit_complex.pdb`).

Cluster packages: `psoriasis_docking_package.tar.gz` (full-scale docking, all 3 targets), `psoriasis_cofolding_package.tar.gz` (Boltz-2 GPU cross-check), `psoriasis_md_package.tar.gz` (equilibrated OpenMM systems).

Figures: volcano, druggability landscape, target ranking, lead structures, pocket druggability, library properties, docking results, top-hit pose, and the pipeline funnel.

Reports: `CONSOLIDATED_REPORT.md` (the narrative synthesis) and this white-paper set (per-step deep dives with references).

4. Reproducibility and provenance

Every figure and table is a tracked artifact with code lineage, and every methodological choice is documented in the per-step white papers. Two caveats are carried consistently through every artifact: the screening library is ChEMBL-bioactivity-sourced with purchasability pending ZINC22 recovery; and STAT3's shallow SH2 pocket makes it a degrader/MD target rather than a classical-docking one.

5. How to run the packaged work

Each cluster package is self-contained (`environment.yml`, prepared inputs, SLURM scripts, README) and runs from a single `sbatch` after extraction: the docking package scales the screen across nodes; the co-folding package runs the Boltz-2 GPU cross-check on the top hits; and the MD package runs production from the pre-equilibrated systems and analyzes the trajectories.

6. Clinical framing

This is research and informational output, not clinical guidance. The leads and compounds identified are computational hypotheses that require experimental validation; therapeutic decisions must be made by qualified professionals with full patient context.

This white-paper set documents each pipeline step with references to the primary literature that established the methods and concepts used. All references are verified against CrossRef.

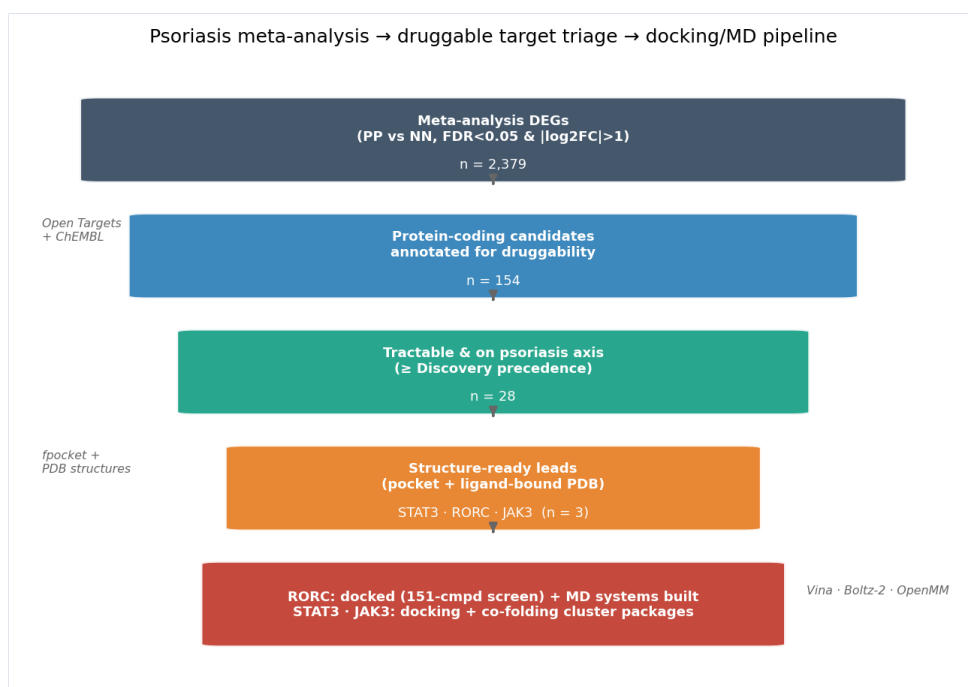


Figure: key output of this pipeline step.

References